IntelliTwin Research Appendix: Authoritative Hypotheses, Metrics, and Evaluation Rules

Derived Rendering of Frozen Research Decisions (ADR-018)

IntelliTwin Research Initiative

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Document Authority Notice: This document represents a derived, human-readable LaTeX rendering of the authoritative hypothesis specifications version-controlled in:

- Primary Specification: `research/hypotheses/hypothesis_framework.yaml`
- Statistical Protocol: `research/research_design/statistical_inference_protocol.md`
- Matched Control Specification: `research/research_design/step09_control_selection.md`
- Pinned Repository Commit: `c163bb279e7427b8f1b6e75955af4969ff9370fd` (Audit Execution: `550a7994d5746d3bcbfa28c26e7ed4f1a26a6153`)

In the event of any textual ambiguity between this rendering and the YAML/protocol files, the version-controlled YAML and statistical protocol remain the primary, immutable scientific authorities.

Governance Rules on Hypothesis Integrity (ADR-018)

Authoritative Precedence: This document is a derived LaTeX rendering. The YAML specification (`hypothesis_framework.yaml`) and statistical protocol (`statistical_inference_p` remain the primary immutable authorities.

Decision Outcome Rules: Each research question possesses a prespecified primary outcome, comparator set, frozen numerical practical-effect threshold, and cluster-bootstrap decision rule classifying results into a three-way outcome: **Supported**, **Unsupported**, or **Inconclusive**.

Outcome Hierarchy: The primary outcome determines the confirmatory claim for each research question. Secondary outcomes verify operational constraints, reliability trade-offs, and failure safeguards. A favorable secondary outcome cannot replace failure on the primary outcome.

No Post-Hoc Alteration: Hypotheses, comparators, subgroups, thresholds, and decision rules are frozen and must not be altered after observing confirmatory test data. Any post-hoc modification must be explicitly designated as an exploratory analysis.

Statistical Sample Unit: The independent statistical unit is the physical **engine unit** ($i \in \{1, \ldots, N\}$). Individual operational cycles (t) within an engine trajectory are autocorrelated and must **NEVER** be treated as independent statistical observations.

Numerical Evaluation Contract (Protocol `step06-v2`)

- **Primary Maintenance Cost Scenario**: `primary_reference` ($C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$ SCU per engine).

- **Secondary Sensitivity Scenarios**: `failure_sensitive` ($C_{\text{FAIL}} = 10.0$) and `waste_sensitive` ($C_{\text{WASTE}} = 0.40$).
- **Nominal Coverage**: Primary confirmatory coverage 90% ($\alpha = 0.10$); descriptive sensitivities at 80% and 95%.
- **Cost Improvement Margin**: $\Delta_{\text{cost}} > 0.20$ SCU per engine (strict inequality).
- **Coverage Error Improvement Margin**: $\Delta_{\text{error}} > 0.01$ (one percentage point error reduction; strict inequality).
- **Maximum Absolute Coverage Error Ceiling**: $E < 0.05$ (prespecified 85%–95% empirical acceptance band).
- **Interval Noninferiority Margins**: $D_{\text{width}} < 1.0$ cycle for MPIW and $D_{\text{score}} < 1.0$ cycle for Winkler interval score.
- **Robustness Non-Harm Margins**: $\Delta_{\text{cost}} > -0.20$ SCU per engine and $\Delta_{\text{error}} > -0.01$ in every cell.
- **Failure Safeguard**: Observed fleet failure rate increase $\Delta_{\text{fail}} \leq 0.0$ against comparator (empirical gate).
- **Asymmetry Calibration**: $\gamma = 2, \beta_{\text{code}} = -1, \beta = -\ln(2) \approx -0.693147$.
- **Fixed Model Seeds**: Five prespecified training seeds: `[2026, 2027, 2028, 2029, 2030]`.

Statistical Resampling and Multiplicity Control

1. **Resampling Engine**: 10,000 paired engine-cluster bootstrap resamples with replacement, stratified by dataset lifetime quartiles (Q_1–Q_4) fixed from split manifests.
2. **PRNG Specification**: `numpy.random.Generator` backed by `PCG64` with seed sequence `SeedSequence([2026, 6, dataset_number])`.
3. **Decision Confidence Level**: Confirmatory decision rules use **98.75% cluster-bootstrap percentile intervals** (quantiles 0.00625 and 0.99375), providing a prespecified nominal Bonferroni family-wise error rate control ($\alpha = 0.05/4 = 0.0125$) across the four primary research questions. Percentile bootstrap inference is approximate and does not guarantee exact finite-sample error control. Descriptive 95% intervals (quantiles 0.025 and 0.975) are reported for context.
4. **Subgroup Families & Bound Propagation**: Operating regime and lifecycle stage subgroup families are evaluated separately but may contain overlapping observations. Subgroup eligibility (≥ 10 engines) is frozen before resampling. If an eligible subgroup is absent in replicate b, its absolute error is bounded by $[0.0, 0.90]$ at 90% nominal coverage, propagating conservatively.
5. **Failure Gate**: The observed failure safeguard ($\Delta_{\text{fail}} \leq 0.0$) is an empirical gate, not a confidence-interval proof of population safety.
6. **Intersection-Union Testing (IUT)**: An RQ is Supported if and only if **every required component condition passes** at the 98.75% decision interval level.

Formal Hypothesis Definitions

Research Question 1 (RQ1): Sequential Maintenance Utility and Coverage Reliability

- **Question**: *Can a decision-aware, condition-adaptive conformal RUL calibration method improve sequential maintenance utility while preserving reliable prediction-interval coverage?*
- **Null Hypothesis ($H_{0,1}$)**: The proposed method does not improve sequential maintenance utility by more than 0.20 SCU per engine over both global and standard condition-aware calibration, or fails to maintain required coverage reliability.
- **Alternative Hypothesis ($H_{1,1}$)**: The proposed method improves maintenance utility by more than 0.20 SCU per engine over both global and standard condition-aware calibration while maintaining required coverage reliability.
- **Primary Outcome**: Total simulated maintenance cost per engine contrast ($\Delta_{\text{cost}} = \overline{\text{Cost}}_B - \overline{\text{Cost}}_P$).
- **Primary Comparators**: Global conformal calibration (`global_hcp`) and standard condition-aware calibration (`regime_hcp`) on C-MAPSS FD002 held-out test split under `primary_reference` at 90% nominal coverage (both GBR and LSTM required).
- **Supported Logic**: For both GBR and LSTM, lower 98.75% CI of $\Delta_{\text{cost}} > 0.20$ SCU against both baselines, upper 98.75% CI of marginal, worst-regime, and worst-stage coverage error < 0.05, and observed fleet failure rate increase ≤ 0.0 against both baselines.
- **Unsupported Logic**: For either model, upper 98.75% CI of $\Delta_{\text{cost}} < 0.20$ SCU against either baseline, lower 98.75% CI of any coverage error > 0.05, or observed fleet failure rate increases over either baseline.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Research Question 2 (RQ2): Conditional Subgroup Reliability and Interval Sharpness

- **Question**: *Can the proposed method improve reliability across operating regimes and degradation conditions without excessively widening prediction intervals?*
- **Null Hypothesis ($H_{0,2}$)**: The proposed method does not improve worst-group conditional coverage error by more than 0.01 coverage-error reduction over both baselines, or causes interval width or interval score to deteriorate by 1.0 cycle or more.
- **Alternative Hypothesis ($H_{1,2}$)**: The proposed method improves worst-group conditional coverage error by more than 0.01 coverage-error reduction over both baselines while restricting interval width and score deterioration to under 1.0 cycle.
- **Primary Outcome**: Worst-group conditional coverage error contrast ($\Delta_{\text{error}} = \text{WorstErr}_B - \text{WorstErr}_P$).
- **Required Subgroup Families**: Operating regimes (C_1–C_6) and degradation lifecycle stages (Early, Mid, Late thirds).
- **Supported Logic**: For both models and both subgroup families, lower 98.75% CI of $\Delta_{\text{error}} > 0.01$ against both baselines, upper 98.75% CI of worst-group error < 0.05, upper 98.75% CI of $D_{\text{width}} < 1.0$ cycle, and upper 98.75% CI of $D_{\text{score}} < 1.0$ cycle.

- **Unsupported Logic**: For either model or subgroup family, upper 98.75% CI of $\Delta_{\text{error}} < 0.01$ against either baseline, lower 98.75% CI of worst-group error > 0.05, lower 98.75% CI of $D_{\text{width}} > 1.0$ cycle, or lower 98.75% CI of $D_{\text{score}} > 1.0$ cycle.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Research Question 3 (RQ3): Decision-Value Attribution vs. Matched Controls

- **Question**: *Are the maintenance benefits of the proposed method attributable to decision-aware uncertainty calibration rather than simply more conservative prediction intervals or earlier maintenance?*
- **Null Hypothesis ($H_{0,3}$)**: The proposed method does not improve maintenance utility by more than 0.20 SCU per engine over every prespecified matched-conservatism control family, or violates coverage constraints.
- **Alternative Hypothesis ($H_{1,3}$)**: The proposed method provides maintenance improvements of more than 0.20 SCU per engine beyond those achievable through matched-conservatism baselines, demonstrating useful decision information rather than simple over-conservatism.
- **Primary Outcome**: Maintenance cost improvement over matched-conservatism controls ($\Delta_{\text{cost}} = \overline{\text{Cost}}_{\text{control}} - \overline{\text{Cost}}_P$).
- **Four Evaluated Control Configurations across Three Families**:
  1. `matched_width_interval`: λ-scaled baseline interval matching proposed validation MPIW within ≤ 1.0 cycle (GBR: `global_hcp`, $\lambda = 0.92$; LSTM: `regime_hcp`, $\lambda = 0.90$).
  2. `widened_interval`: Parameterized interval expansion $\lambda \geq 1.0$ (GBR: `regime_hcp`, $\lambda = 1.00$; LSTM: `global_hcp`, $\lambda = 1.00$). Note: $\lambda = 1.00$ does not represent actual widening.
  3. `point_safety_margin`: Thresholding $p \leq 1 + m$ (GBR: $m = 5$; LSTM: $m = 4$).
  4. `earlier_intervention`: Elevated thresholding $L \leq h$ (GBR: `regime_hcp`, $h = 1$; LSTM: `global_hcp`, $h = 1$). Note: $h = 1$ represents baseline thresholding, not an empirically earlier intervention.
- **Supported Logic**: For both GBR and LSTM, lower 98.75% CI of $\Delta_{\text{cost}} > 0.20$ SCU against **every single prespecified control configuration**, upper 98.75% CI of marginal, regime, and stage coverage error < 0.05, and observed fleet failure rate increase ≤ 0.0 against all controls.
- **Unsupported Logic**: For either model, upper 98.75% CI of $\Delta_{\text{cost}} < 0.20$ SCU against any control configuration, lower 98.75% CI of any coverage error > 0.05, or fleet failure rate increases over any control.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Research Question 4 (RQ4): Robustness and Generalization

- **Question**: *Does the proposed method retain its calibration and maintenance-decision benefits across operating conditions, base RUL predictors, datasets, and repeated experimental runs?*

- **Null Hypothesis ($H_{0,4}$)**: The proposed method fails to achieve aggregate improvements ($\Delta_{\text{cost}} > 0.20$ SCU, $\Delta_{\text{error}} > 0.01$) across the evaluation matrix, or exhibits material harm ($\Delta_{\text{cost}} < -0.20$ SCU or $\Delta_{\text{error}} < -0.01$) in any individual configuration cell.
- **Alternative Hypothesis ($H_{1,4}$)**: The proposed method achieves aggregate improvements across the evaluation matrix without material harm in any individual configuration cell.
- **Evaluation Matrix**: Six mandatory cells: FD001 × GBR, FD001 × LSTM, FD002 × GBR, FD002 × LSTM, FD004 × GBR, FD004 × LSTM, weighted equally at $w_k = 1/6$ per cell.
- **Supported Logic**: Lower 98.75% CI of weighted aggregate $\Delta_{\text{cost}} > 0.20$ SCU, lower 98.75% CI of weighted aggregate $\Delta_{\text{error}} > 0.01$ for regimes and stages, lower 98.75% CI of $\Delta_{\text{cost}} > -0.20$ SCU in every cell, lower 98.75% CI of $\Delta_{\text{error}} > -0.01$ in every cell, and all cell safeguards hold across all 6 cells.
- **Unsupported Logic**: Upper 98.75% CI of aggregate $\Delta_{\text{cost}} < 0.20$ SCU or aggregate $\Delta_{\text{error}} < 0.01$, or upper 98.75% CI of cell $\Delta_{\text{cost}} < -0.20$ SCU or $\Delta_{\text{error}} < -0.01$ in any individual cell, or any cell safeguard fails.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Summary Evaluation Contract Matrix

ID	Experiment	Primary Outcome	Key Comparators	Statistical Unit	Practical Threshold (98.75% CI)
RQ1	Experiment A	Maintenance cost per engine (Δ_{cost})	`global_hcp` & `regime_hcp`	Engine ($N = 52$)	$\Delta_{\text{cost}} > 0.20$ SCU per engine
RQ2	Experiment B	Worst-group coverage error (Δ_{error})	`global_hcp` & `regime_hcp`	Engine ($N = 52$)	$\Delta_{\text{error}} > 0.01$; $D_{\text{width}}, D_{\text{score}} < 1.0$
RQ3	Experiment C	Utility vs. matched controls (Δ_{cost})	4 named matched controls	Engine ($N = 52$)	$\Delta_{\text{cost}} > 0.20$ SCU vs. all controls
RQ4	Experiment D	Weighted aggregate & cell non-harm	6 cells ($w_k = 1/6$ each)	Engine ($N = 122$)	Agg: $> 0.20, > 0.01$; Harm: $> -0.20, > -0.01$

Table 1: IntelliTwin Confirmatory Hypotheses, Metrics, Comparators, and Frozen Thresholds (ADR-018).